

Ming Li

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Research Profile

Ph.D. student in Computer Science at the University of Maryland, advised by Prof. Tianyi Zhou. My research develops post-training, interpretability, and evaluation methods to understand and improve the behavior of large language models and agentic systems, with a focus on data-centric adaptation, reasoning behavior analysis, and agent evaluation. I have led 10+ first-author papers at ACL, NeurIPS, ICLR, EMNLP, and NAACL, served as an *ACL ARR Area Chair*, and received the **2026 Apple PhD Scholar in AI/ML Fellowship**.

Education

University of Maryland, Ph.D. in Computer Science	Advisor: Prof. Tianyi Zhou	Aug 2023 – Present
Texas A&M University, M.S. in Computer Science	Advisor: Prof. Ruihong Huang	Aug 2021 – May 2023
Xi'an Jiaotong University, B.S. Computer Science		Aug 2016 – June 2020

Internship & Research Experience

MBZUAI, Visiting Researcher	Feb 2026 – May 2026
- Studied agent evaluation and AI socialization in multi-agent environments. - Developed evaluation protocols for coordination, social behavior, and emergent interaction among LLM agents.	
Amazon, Applied Scientist Intern	May 2025 – Aug 2025
- Developed reinforcement learning methods and evaluation pipelines for optimizing multi-turn LLM conversations. - Studied reward design, trajectory-level feedback, and conversational behavior under iterative post-training.	
Microsoft Research, Research Intern	Feb 2025 – May 2025
- Studied interpretability and collaborative decoding mechanisms for LLM reasoning. - Analyzed how decoding strategies affect reasoning quality, efficiency, and model behavior.	
Adobe, Research Scientist/Engineer Intern	May 2024 – Nov 2024
- Developed visual grounding and instruction-tuning methods for multimodal LLMs. - Built data and evaluation pipelines for improving grounded instruction following in MLLMs.	

Research Areas

- Data-centric AI and Post-training** [1–4]: Developed data-centric and algorithmic approaches for improving LLM adaptation, instruction following, and reasoning, including data selection, synthesis, reflection tuning, and data recycling.
- Model Behavior and Interpretability** [5–8]: Analyzed how training data, post-training objectives, inference strategies, and interaction environments shape the behaviors of LLMs and agents, including reasoning, overthinking, and collaboration.
- Agentic Systems, Benchmarks, and Evaluation Harnesses** [9–12]: Built benchmarks, methods, and environments for evaluating VLM capabilities, LLM agents, multi-agent societies, and agent-environment interaction.

Selected Publications

- [1] Li, M., Zhang, Y., Li, Z., et al. From quantity to quality: Boosting LLM performance with self-guided data selection for instruction tuning. *NAACL'24*.
- [2] Li, M., Zhang, Y., He, S., et al. Superfiltering: Weak-to-strong data filtering for fast instruction-tuning. *ACL'24*.
- [3] Li, M., Chen, L., Chen, J., He, S., Gu, J., & Zhou, T. Selective reflection-tuning: Student-selected data recycling for LLM instruction-tuning. *ACL'24*.
- [4] Li, M., Li, Y., Li, Z., & Zhou, T. How instruction and reasoning data shape post-training: Data quality through the lens of layer-wise gradients. *ACL'26*.
- [5] Li, M., Fan, C., Cheng, Y., Feizi, S., & Zhou, T. Schoenfeld's anatomy of mathematical reasoning by language models. *ACL'26, Oral, SAC Highlight*.
- [6] Li, M., et al. What happened in LLMs layers when trained for fast vs. slow thinking: A gradient perspective. *ACL'25, Oral*.
- [7] Fan, C.*, Li, M.*, Sun, L., & Zhou, T. Missing premise exacerbates overthinking: Are reasoning models losing critical thinking skill? *COLM'25*.
- [8] Li, X.*, Li, M.*, & Zhou, T. What does RL improve for visual reasoning? A Frankenstein-style analysis. *COLM'26*.
- [9] Liang, Y.*, Li, M.*, Fan, C., et al. ColorBench: Can VLMs see and understand the colorful world? *NeurIPS'25*.
- [10] Chen, Z.*, Li, M.*, Huang, Y., et al. ATLaS: Agent tuning via learning critical steps. *ACL'25*.
- [11] Li, M., Li, X., & Zhou, T. Does socialization emerge in AI agent society? A case study of Moltbook. *ACM CAIS'26*.
- [12] Li, X., Li, M., et al. ClawEnvKit: Automatic environment generation for claw-like agents.

Honors, Service, Talks, and Open Source

- Honors:** **2026 Apple PhD Scholar in AI/ML Fellowship**; **ACL'26 SAC Highlight**; **Best Research Award at e-Assessment Awards 2026**; UMD Graduate School Dean's Fellowship
- Academic Service:** ACL ARR Area Chair; reviewer for ACL, EMNLP, NAACL, ICLR, ICML, and NeurIPS
- Talks, Tutorials, and Media:** ACM CIKM 2025 Tutorial on Data Synthesis; *Science News* interview on AI Agent societies
- Open Source:** Cherry_LLM, Reflection_Tuning, Superfiltering, and other post-training projects, with 1000+ combined GitHub stars